

Conceptual Design Document Rev 00 — Gen-3 Safe Helicopter / Heavy Modular Amphibious VTOL Platform

Subtitle: Gen-3A 100-ton Pure Rotorcraft and Gen-3B 200-ton Hybrid Buoyant Rotorcraft

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Revision: 00

Status: Concept Baseline / Pre-Feasibility

Classification: Proprietary Concept Information

1. LEGAL NOTICE AND CAVEATS

All numerical values contained herein are conceptual design targets and require rigorous validation by CFD, FEA, aeroelastic analysis, system safety assessment, full-scale testing, and certification authorities. This document does not claim certified safety, airworthiness, manufacturability, or guaranteed performance.

2. DOCUMENT PURPOSE

This document consolidates the conceptual baseline for two Generation-3 safety-centered modular VTOL aircraft. "100-ton" and "200-ton" refer to target maximum operational mass baselines. At Rev 00, a complete aircraft design is not achievable; this document serves as a comprehensive conceptual requirements framework, explicitly identifying unresolved engineering items for future iteration.

3. MANDATORY COMMON REQUIREMENTS

- Amphibious Capability: Both platforms are designed for controlled water landing, flotation, and water takeoff within a defined sea-state envelope.
 - Safety Priority: Engineering targets include the mitigation of fire, explosion, electrocution, structural breakup, and fuel release during survivable water impact. Crashworthiness reduces risk but cannot guarantee total suppression of explosion in extreme impact events.
 - Structural Composition: Load-bearing airframe and skin structural subsystems target 80% carbon-fiber-composite by mass. Metallic joints, lightning protection, firewalls, and impact structures remain integral.
 - Safety Capsule: A centrally located, rapidly detachable, independently buoyant capsule for 30 passengers and crew. Features include 72-hour life support, emergency oxygen objectives, GNSS/distress beacons, and flotation.
 - Parachute System: Exactly 12 main canopies arranged in four independent modules of three canopies each, featuring sequenced deployment, reefing, and entanglement mitigation.
 - Capsule Propulsion: Independent downward-facing/underslung ducted fans for attitude control and descent-rate reduction.
 - Modularity & Jettison: Central capsule, peripheral fans, rotor arms, and turbogenerator modules are modular. Fire response: Detect → Isolate → Suppress → Stabilize → Jettison (only if necessary). Jettison is subject to strict flight-control authority and safe-corridor verification.
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4. DESIGN CRITERIA AND FORMULAS

The conceptual target thrust-to-weight ratio is $T/W = 1.7$. Basic performance relations:

$$T_{target} = 1.7W$$

$$A_{disk} = \pi(D/2)^2$$

$$P_i = \frac{T^{3/2}}{2\rho A}$$

5. GEN-3A: 100-TON PURE ROTORCRAFT

- Operational Mass: 100 t
 - Total Thrust: 170 tf (~1.667 MN)
 - Hover Power: 33.77 MW (Baseline); 46.90 MW (Design)
 - Installed Generation: 55–60 MW (8 × 7.5 MW turbogenerators)
 - Lift Distribution: 12 peripheral fans (12 m dia.) + 4 underbody rotors (10 m dia.)
 - Preliminary Mass Ledger:
 - Composite Structure/Skin: 25 t
 - Fans/Arms/Ducts: 18 t
 - Motors/Inverters: 10 t
 - Turbogenerators/Fuel Delivery: 15 t
 - Safety Capsule: 10 t
 - Avionics/Controls: 5 t
 - Landing/Flotation: 5 t
 - Fuel: 7 t
 - Occupants/Supplies: 5 t
 - Note: Requires 15–25% development margin in future iterations.
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6. GEN-3B: 200-TON HYBRID BUOYANT ROTORCRAFT

- Operational Mass: 200 t
 - Helium Volume: 119,389 m³ (Target)
 - Gross Buoyancy: $L_b = (\rho_{air} - \rho_{He})gV$
 - Thrust Requirement: 340 tf (~3.334 MN) to minimize reliance on buoyancy.
 - Hover Power: 78.48 MW (Baseline); 108.99 MW (Design)
 - Installed Generation: 125–130 MW (10 × 13 MW turbogenerators)
 - Lift Distribution: 16 peripheral fans (12 m dia.) + 8 underbody rotors (10 m dia.)
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7. AMPHIBIOUS AND SAFETY ARCHITECTURE

- Hydrodynamic Hull: Trimaran/catamaran-inspired with crushable keel.

- Water Impact: High-voltage systems require automatic contactor opening and bus segmentation.
- Fuel Safety: Self-sealing, crashworthy bladders in protected zones; no tanks in peripheral modules.
- Capsule Separation: Initiated only after mother-aircraft isolation; requires clearance from debris/rotor paths.

8. REQUIREMENTS AND VERIFICATION MATRIX

ID	Requirement	Method	Status
G3-COM-001	1.7 T/W Ratio	Analysis	Open
G3-AMP-001	Water-Impact Sealing	Test	Open
G3-CAP-001	12-Canopy Deployment	Simulation	Open
G3-MOD-001	Fire Suppression/Jettison	Analysis	Open

9. RISK REGISTER (HIGH-LEVEL)

Risk	Likelihood	Consequence
Installed-power mass/volume	High	High
Rotor/Fan Interference	High	High
Capsule deployment envelope	Medium	Critical
Water-takeoff ingestion	Medium	High

10. VALIDATION ROADMAP

1. Phases 0-1: Parametric Sizing & Mission Definition.
 2. Phases 2-4: CFD/FEA & Hardware-in-Loop Control Simulation.
 3. Phases 5-6: Subscale testing, model-basin hydrodynamic trials, capsule drop tests.
 4. Phases 7-8: Unmanned flight demonstrator followed by certification-basis finalization.
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DOCUMENT CONTROL

Founder: Amirbahador Atai

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Certification Status: Not certified